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CSCD 240

Winter 2019

Lab 2: More Unix Commands

‘echo’

1. What are the differences among the following commands? Explain with screenshots.
(6 pts , 1 point each)

   a) echo cal
   will just print “cal” to the screen

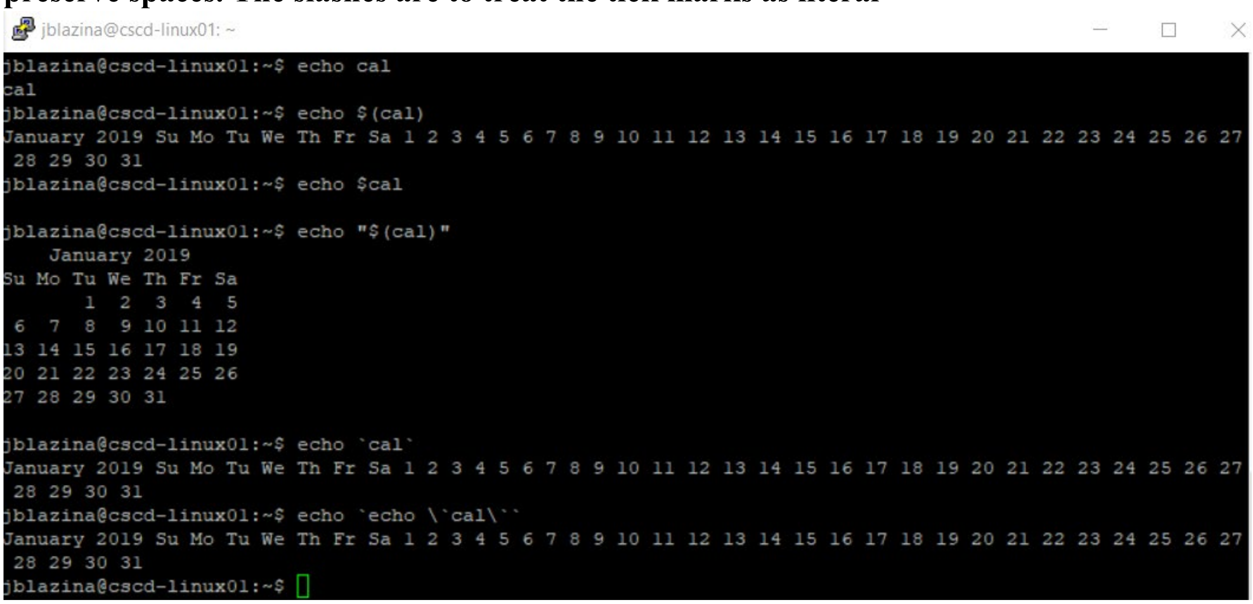
   b) echo $(cal)
   will print out an unformatted current month calendar to the screen. Does not preserve spaces

   c) echo $cal
   will try to print out a variable named cal. If there is no variable it will print nothing

   d) echo “$(cal)”
   will print out a formatted current month calender to the screen. Preserves spaces.

   e) echo `cal` ←This is back quote
   will print out an unformatted current current month calendar to the screen. Does not preserve spaces.

   f) echo `echo \`cal\`` ← This is back quote
   will print out an unformatted current month calendar to the screen. Does not preserve spaces. The slashes are to treat the tick marks as literal



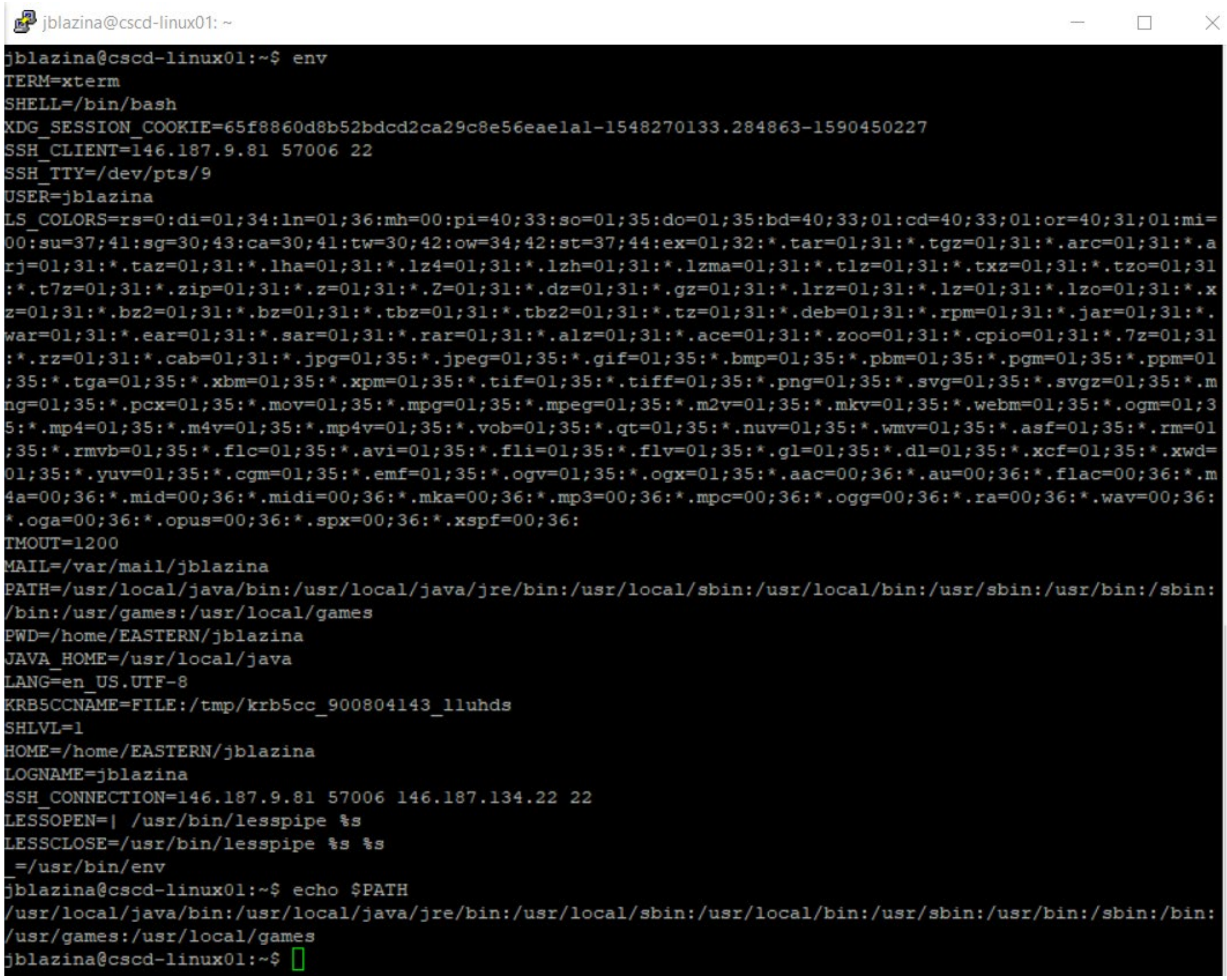
Environment variables

2. What command will show you all the environment variables? What command will display the environment variable named **PATH**? Show both with screenshot.

(2 points, 1 point each)

   a) env

   b) echo $PATH



```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ env
TERM=xterm
SHELL=/bin/bash
XDG_SESSION_COOKIE=65f8860d8b52bdcd2ca29c8e56eae1a1-1548270133.284863-1590450227
SSH_CLIENT=146.187.9.81 57006 22
SSH_TTY=/dev/pts/9
USER=jblazina
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=
00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.a
rj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31
:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.Z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.x
z=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.
war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31
:*.rz=01;31:*.cab=01;31:*.jpg=01;35:*.jpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01
;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.m
ng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;3
5:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01
;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=
01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m
4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:
*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
TMOUT=1200
MAIL=/var/mail/jblazina
PATH=/usr/local/java/bin:/usr/local/java/jre/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:
/bin:/usr/games:/usr/local/games
PWD=/home/EASTERN/jblazina
JAVA_HOME=/usr/local/java
LANG=en_US.UTF-8
KRB5CCNAME=FILE:/tmp/krb5cc_900804143_1luhds
SHLVL=1
HOME=/home/EASTERN/jblazina
LOGNAME=jblazina
SSH_CONNECTION=146.187.9.81 57006 146.187.134.22 22
LESSOPEN=| /usr/bin/lesspipe %s
LESSCLOSE=/usr/bin/lesspipe %s %s
_=/usr/bin/env
jblazina@cscd-linux01:~$ echo $PATH
/usr/local/java/bin:/usr/local/java/jre/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:
/usr/games:/usr/local/games
jblazina@cscd-linux01:~$
```

I/O

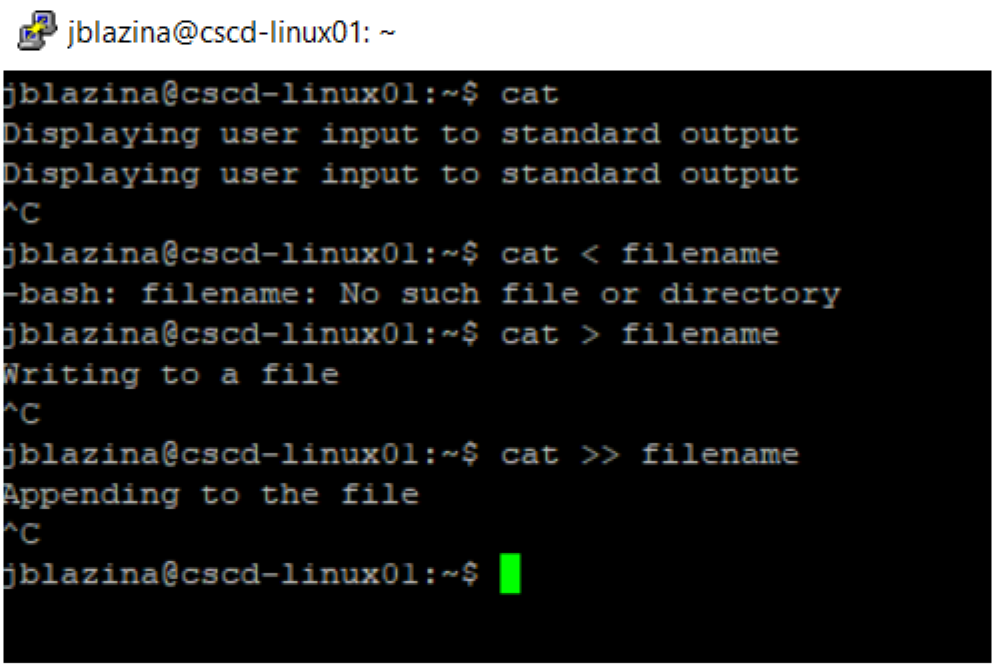
3. What are the differences among the following commands. Explain with examples and screenshot. (4 points, 1 point each)

cat **Displays user input to the standard output**

cat < filename **Reading from filename and displaying to standard output**

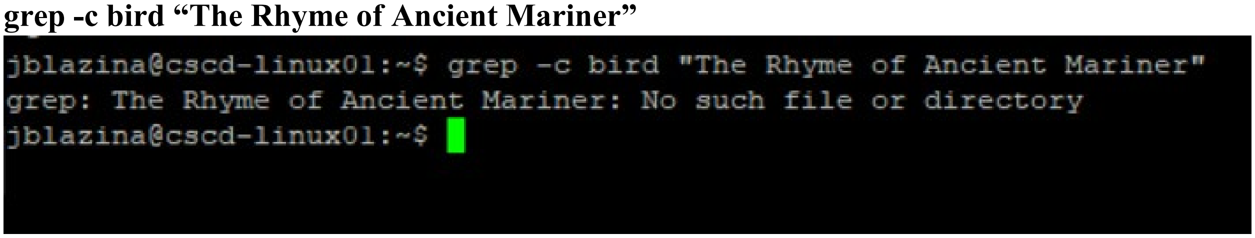
cat > filename **Redirects user input to filename and overwrites the content or creates a file if none exists**

cat >> filename **Redirects user input to filename and appends to the existing content**



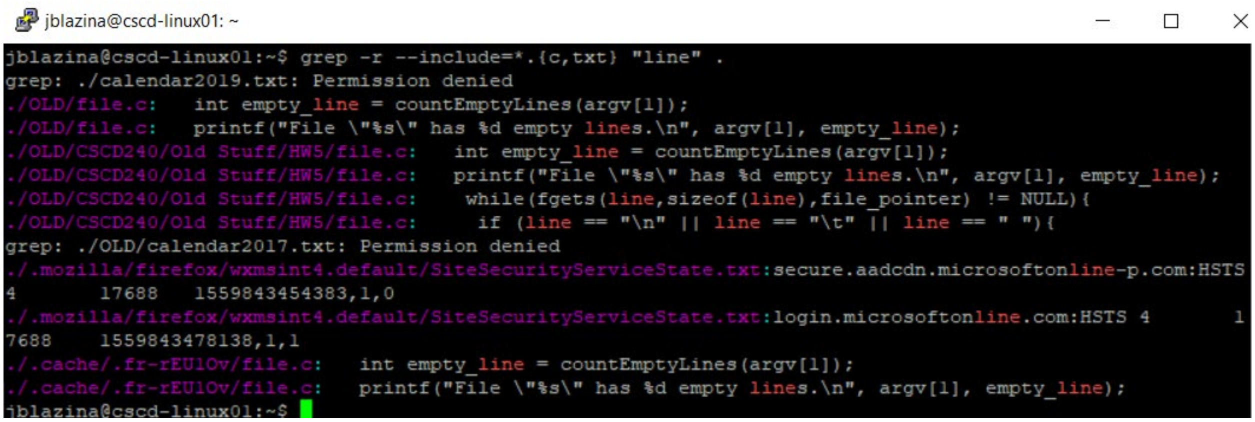
```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ cat
Displaying user input to standard output
Displaying user input to standard output
^C
jblazina@cscd-linux01:~$ cat < filename
-bash: filename: No such file or directory
jblazina@cscd-linux01:~$ cat > filename
Writing to a file
^C
jblazina@cscd-linux01:~$ cat >> filename
Appending to the file
^C
jblazina@cscd-linux01:~$
```

4. Write a command that counts the total number of lines the string "**bird**" exists in a file named "**The Rhyme of Ancient Mariner**" in your current directory. (1 point)
grep -c bird "The Rhyme of Ancient Mariner"



```
jblazina@cscd-linux01:~$ grep -c bird "The Rhyme of Ancient Mariner"
grep: The Rhyme of Ancient Mariner: No such file or directory
jblazina@cscd-linux01:~$
```

5. Write a command that searches the string "**line**" in all .c and .txt files starting from your current directory and all sub directories. (1 point)
grep -r –include=*.{c,txt} "line" .
note: there are two dashes in front of include



```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ grep -r --include=*.{c,txt} "line" .
grep: ./calendar2019.txt: Permission denied
./OLD/file.c:    int empty_line = countEmptyLines(argv[1]);
./OLD/file.c:    printf("File \"%s\" has %d empty lines.\n", argv[1], empty_line);
./OLD/CSCD240/Old Stuff/HW5/file.c:    int empty_line = countEmptyLines(argv[1]);
./OLD/CSCD240/Old Stuff/HW5/file.c:    printf("File \"%s\" has %d empty lines.\n", argv[1], empty_line);
./OLD/CSCD240/Old Stuff/HW5/file.c:     while(fgets(line,sizeof(line),file_pointer) != NULL){
./OLD/CSCD240/Old Stuff/HW5/file.c:      if (line == "\n" || line == "\t" || line == " "){
grep: ./OLD/calendar2017.txt: Permission denied
./.mozilla/firefox/wxmsint4.default/SiteSecurityServiceState.txt:secure.aadcdn.microsoftonline-p.com:HSTS
4       17688    1559843454383,1,0
./.mozilla/firefox/wxmsint4.default/SiteSecurityServiceState.txt:login.microsoftonline.com:HSTS 4       1
7688    1559843478138,1,1
./.cache/.fr-rEU1Ov/file.c:    int empty_line = countEmptyLines(argv[1]);
./.cache/.fr-rEU1Ov/file.c:    printf("File \"%s\" has %d empty lines.\n", argv[1], empty_line);
jblazina@cscd-linux01:~$
```

Metacharacters in Regular Expression

6. What will the following patterns match? Explain. (4 points, 1 point each)
   a) ^bags$
   Will look specifically for bags because it must start with bags and end with bags
   b) ^…$
   Will look for anything with three characters
   c) l.g
   Will look for any three letter word that starts with l and ends with g. The middle character can be anything
   d) ^\.
   Will look for any line that starts with a period. The slash will treat the . literally.

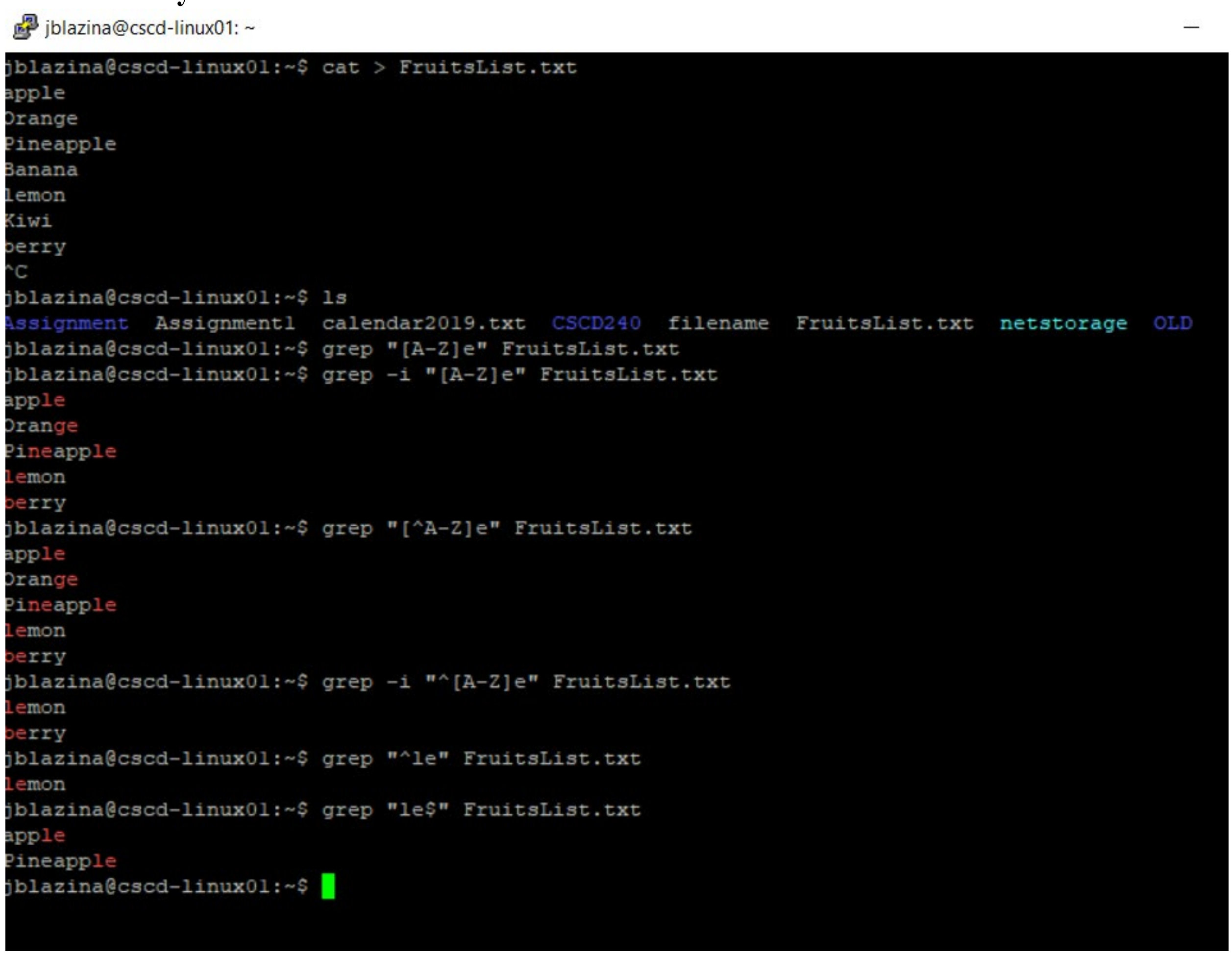
‘grep’, ‘find’, pipe

7. Consider the following file named “**FruitsList.txt**”. Try the following commands and explain each output with screenshot. (6 points, 1 point each)

FruitsList.txt

```
apple
Orange
Pineapple
Banana
lemon
Kiwi
berry
```

a) grep “[A-Z]e” FruitsList.txt
   Looks for a capital letter followed by ‘e’
b) grep -i “[A-Z]e” FruitsList.txt
   Ignores case and looks for a capital letter Followed by ‘e’
c) grep “[^A-Z]e” FruitsList.txt
   Looks for any word that has ‘e’ preceded By a lowercase letter. The ^ says NOT
d) grep –i “^[A-Z]e” FruitsList.txt
   Ignores case and looks for any letter that will be followed by ‘e’.
e) grep “^le” FruitsList.txt
   Looks for any word that starts with ‘le’
f) grep “le$” FruitsList.txt
   Looks for any word that ends with ‘le’



```
jblazina@cscd-linux01:~$ cat > FruitsList.txt
apple
Orange
Pineapple
Banana
lemon
Kiwi
berry
^C
jblazina@cscd-linux01:~$ ls
Assignment  Assignment1  calendar2019.txt  CSCD240  filename  FruitsList.txt  netstorage  OLD
jblazina@cscd-linux01:~$ grep "[A-Z]e" FruitsList.txt
jblazina@cscd-linux01:~$ grep -i "[A-Z]e" FruitsList.txt
apple
Orange
Pineapple
lemon
berry
jblazina@cscd-linux01:~$ grep "[^A-Z]e" FruitsList.txt
apple
Orange
Pineapple
lemon
berry
jblazina@cscd-linux01:~$ grep -i "^[A-Z]e" FruitsList.txt
lemon
berry
jblazina@cscd-linux01:~$ grep "^le" FruitsList.txt
lemon
jblazina@cscd-linux01:~$ grep "le$" FruitsList.txt
apple
Pineapple
jblazina@cscd-linux01:~$
```

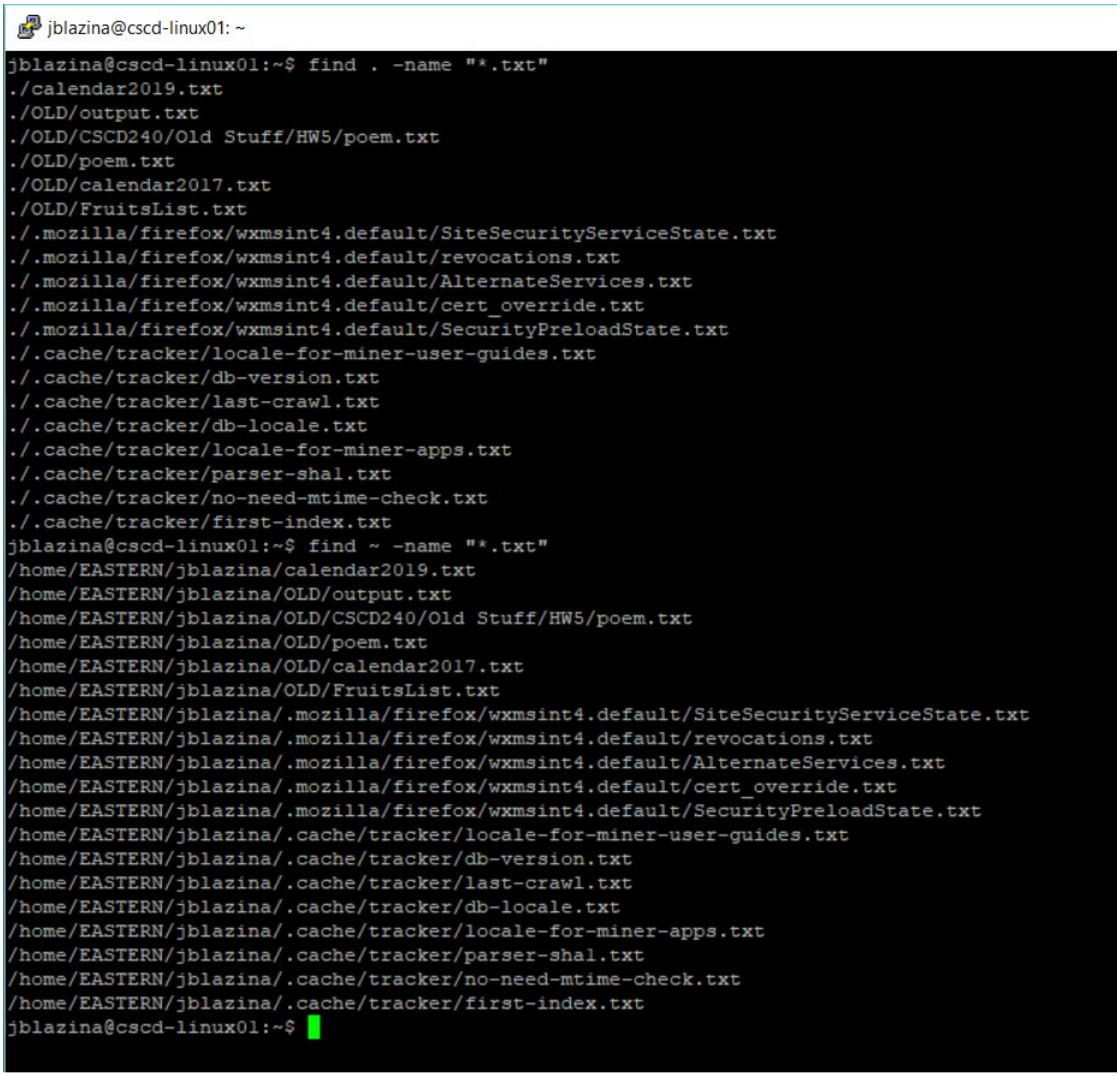
8. Suppose you are in your home directory. What are the differences between the following commands? Explain with screenshot. 1 point

   find . –name "*.txt"

   Any .txt file starting in the current directory and shows the relative path. Find is recursive by default

   find ~ -name "*.txt"

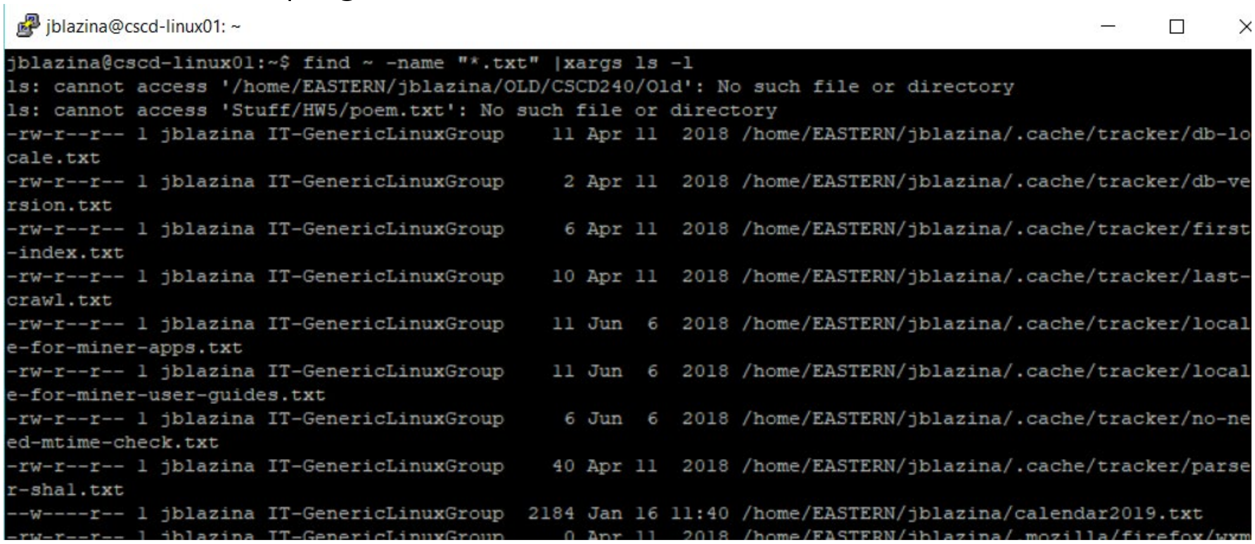
   Any .txt file in the home directory and shows the absolute path



```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ find . -name "*.txt"
./calendar2019.txt
./OLD/output.txt
./OLD/CSCD240/Old Stuff/HW5/poem.txt
./OLD/poem.txt
./OLD/calendar2017.txt
./OLD/FruitsList.txt
./.mozilla/firefox/wxmsint4.default/SiteSecurityServiceState.txt
./.mozilla/firefox/wxmsint4.default/revocations.txt
./.mozilla/firefox/wxmsint4.default/AlternateServices.txt
./.mozilla/firefox/wxmsint4.default/cert_override.txt
./.mozilla/firefox/wxmsint4.default/SecurityPreloadState.txt
./.cache/tracker/locale-for-miner-user-guides.txt
./.cache/tracker/db-version.txt
./.cache/tracker/last-crawl.txt
./.cache/tracker/db-locale.txt
./.cache/tracker/locale-for-miner-apps.txt
./.cache/tracker/parser-shal.txt
./.cache/tracker/no-need-mtime-check.txt
./.cache/tracker/first-index.txt
jblazina@cscd-linux01:~$ find ~ -name "*.txt"
/home/EASTERN/jblazina/calendar2019.txt
/home/EASTERN/jblazina/OLD/output.txt
/home/EASTERN/jblazina/OLD/CSCD240/Old Stuff/HW5/poem.txt
/home/EASTERN/jblazina/OLD/poem.txt
/home/EASTERN/jblazina/OLD/calendar2017.txt
/home/EASTERN/jblazina/OLD/FruitsList.txt
/home/EASTERN/jblazina/.mozilla/firefox/wxmsint4.default/SiteSecurityServiceState.txt
/home/EASTERN/jblazina/.mozilla/firefox/wxmsint4.default/revocations.txt
/home/EASTERN/jblazina/.mozilla/firefox/wxmsint4.default/AlternateServices.txt
/home/EASTERN/jblazina/.mozilla/firefox/wxmsint4.default/cert_override.txt
/home/EASTERN/jblazina/.mozilla/firefox/wxmsint4.default/SecurityPreloadState.txt
/home/EASTERN/jblazina/.cache/tracker/locale-for-miner-user-guides.txt
/home/EASTERN/jblazina/.cache/tracker/db-version.txt
/home/EASTERN/jblazina/.cache/tracker/last-crawl.txt
/home/EASTERN/jblazina/.cache/tracker/db-locale.txt
/home/EASTERN/jblazina/.cache/tracker/locale-for-miner-apps.txt
/home/EASTERN/jblazina/.cache/tracker/parser-shal.txt
/home/EASTERN/jblazina/.cache/tracker/no-need-mtime-check.txt
/home/EASTERN/jblazina/.cache/tracker/first-index.txt
jblazina@cscd-linux01:~$
```

9. Write a command that finds all text files in your home directory and subdirectory and shows the long listing. 1 point

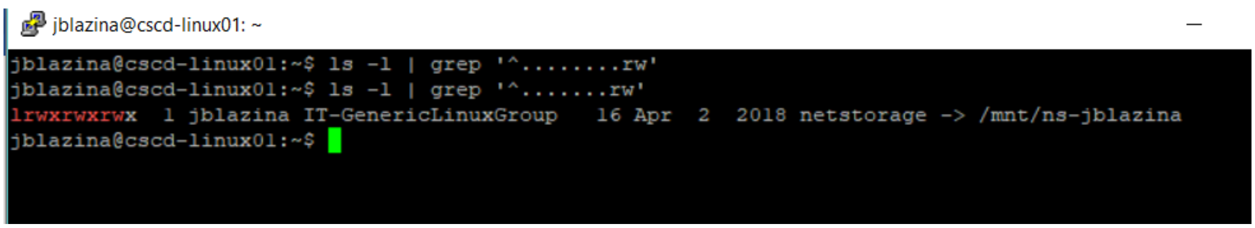
find ~ -name "*.txt" |xargs ls -l



10. What will the following commands do? Explain with screenshots. (2 points, 1 point each)

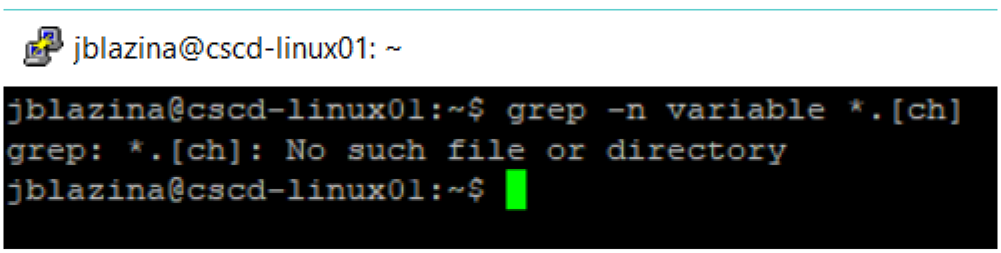
a) ls –l | grep '^…….rw'

Does a long listing and looks for files that have read/write permission for 'others'



b) grep –n variable *.[ch]

Looks for the pattern 'variable' in any files that are .c or .h files and will print the line number

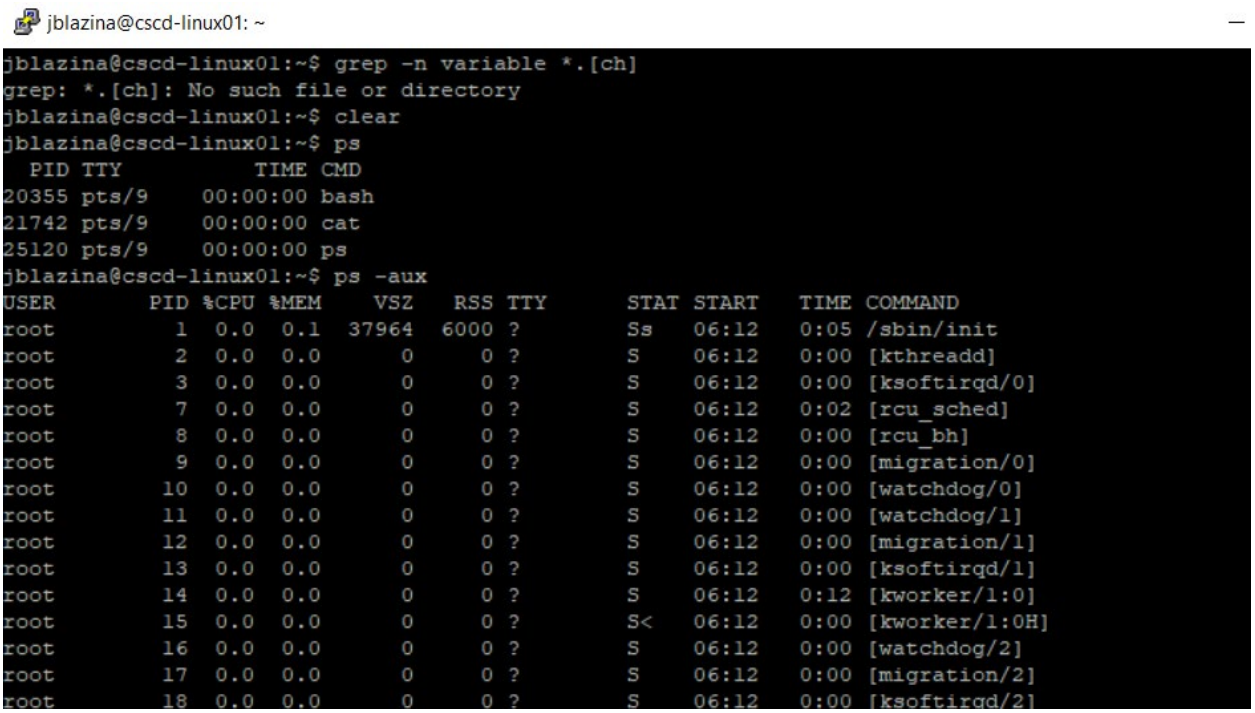


Processes and Jobs

11. What is process? How will you differentiate processes from jobs? 1 point
    Processes are global, including anything that is running in your current shell, a job is something that is bound to your current shell
12. What are the difference between the following commands: Explain with screenshot.
    ps and **ps –aux.**

    1 point

    ps shows all processes for current shell, ps -aux shows all processes in a global scope.



```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ grep -n variable *.[ch]
grep: *.[ch]: No such file or directory
jblazina@cscd-linux01:~$ clear
jblazina@cscd-linux01:~$ ps
  PID TTY          TIME CMD
20355 pts/9    00:00:00 bash
21742 pts/9    00:00:00 cat
25120 pts/9    00:00:00 ps
jblazina@cscd-linux01:~$ ps -aux
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1  37964  6000 ?        Ss   06:12   0:05 /sbin/init
root         2  0.0  0.0      0     0 ?        S    06:12   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    06:12   0:00 [ksoftirqd/0]
root         7  0.0  0.0      0     0 ?        S    06:12   0:02 [rcu_sched]
root         8  0.0  0.0      0     0 ?        S    06:12   0:00 [rcu_bh]
root         9  0.0  0.0      0     0 ?        S    06:12   0:00 [migration/0]
root        10  0.0  0.0      0     0 ?        S    06:12   0:00 [watchdog/0]
root        11  0.0  0.0      0     0 ?        S    06:12   0:00 [watchdog/1]
root        12  0.0  0.0      0     0 ?        S    06:12   0:00 [migration/1]
root        13  0.0  0.0      0     0 ?        S    06:12   0:00 [ksoftirqd/1]
root        14  0.0  0.0      0     0 ?        S    06:12   0:12 [kworker/1:0]
root        15  0.0  0.0      0     0 ?        S<   06:12   0:00 [kworker/1:0H]
root        16  0.0  0.0      0     0 ?        S    06:12   0:00 [watchdog/2]
root        17  0.0  0.0      0     0 ?        S    06:12   0:00 [migration/2]
root        18  0.0  0.0      0     0 ?        S    06:12   0:00 [ksoftirqd/2]
```

<u>Submission:</u>

- A PDF file - Name this file as follows: your last name, first letter of your first name, Lab2.pdf (i.e., YasminSLab2.pdf). This file will contain all your answers. Each question should be copied first and then answered.
- You should turn in through the EWU Canvas system.
- Submission deadline is **Wednesday, January 23**. **No late submission will be accepted**.